MIL-G-22997A(SHIPS)
30 April 1965
SUPERSEDING
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(See 6.3)

MILITARY SPECIFICATION

GENERATOR SET, DIESEL ENGINE,

100 KW, 450-VOLT A.C., 60-CYCLE, 1800 R.P.M.

1. SCOPE

1.1 This specification covers 100 kilowatt (kw.), 450-volt alternating current (a.c.), lightweight, low magnetic signature diesel engine generator sets for Naval shipboard use.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-B-171 - Bearings, Ball, Annular (General Purpose).

MILITARY

MIL-B-857 - Bolts, Nuts, and Studs.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-917 - Electric Power Equipment, Basic Requirements for (Naval Shipboard Use).
MIL-D-963 - Drawings, Electrical, Hull and Mechanical Equipment for Naval Shipboard Use.
MIL-V-1137 - Varnish, Electrical-Insulating (for Electro-Motive Equipment).
MIL-R-2729 - Regulator Set, Voltage, A.C., Generator, (Naval Shipboard Use).
MIL-G-3048 - Generator Set, Diesel Engine, Direct- and -Alternating-Current, (Naval Shipboard Use).
MIL-G-3124 - Generator, Alternating Current, 60-Cycle (Naval Shipboard Use).
MIL-L-9000 - Lubricating Oil, Internal Combustion Engine, Diesel.
MIL-M-15071 - Manuals, Equipment and Systems.
MIL-E-15090 - Enamel, Equipment, Light Gray (Formula No. 11).
MIL-P-15137 - Provisioning and Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-P-15328 - Primer, Pretreatment (Formula No. 117 for Metals).
MIL-M-15337 - Mufflers, Exhaust, Internal Combustion Engine.
MIL-C-15730 - Coolers, Fluid, Industrial, Naval Shipboard, Lubricating Oil, Hydraulic Oil, and Fresh Water.
MIL-S-16032 - Switches, Pressure and Thermostatic, Naval Shipboard Alarm System.
MIL-T-16049 - Tachometer, Electrical, Self-Generating, Mechanical, Fixed Mounting and Hand Held, and Vibrating Reed.
MIL-F-16884 - Fuel Oil, Diesel, Marine.
MIL-I-16910 - Interference Measurement, Electromagnetic, Methods and Limits.
MIL-M-17185 - Mounts, Resilient, General Specifications and Tests for, Shipboard Application.
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts; Preparation for Delivery of.
MIL-F-20627 - Filter Assembly and Filter Elements, Fluid, Pressure, for Engines with Liquid Fuel Injection Systems.
MIL-F-20707 - Filter Elements, Fluid, Pressure, Oil, Full Flow.
MIL-C-21121 - Coolers, Fluid, Industrial, Lubricating Oil and Fresh Water, Naval Shipboard.
MIL-N-25027 - Nut, Self-Locking, 250 Degrees F., 450 Degrees F., and 800 Degrees F., 125 KSI FTU, and 30 KSI FTU.

FSC 6115

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STANDARDS

MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.
MIL-STD-701 - Preferred and Guidance Lists of Semiconductor Devices.
MS17828 - Nuts, Self Locking, Nylon Insert, Hexagon, Regular Height, 250 Degrees F., Nickel-Copper Alloy.
MS17829 - Nuts, Self Locking, Nylon Insert, Hexagon, Regular Height, 250 Degrees F., Grade 8 Carbon Steel.
MS17830 - Nuts, Self Locking, Plastic Insert, Hexagon, Regular Height, 250 Degrees F., FS303 Corrosion Resistant Steel.

PUBLICATIONS

BUREAU OF SHIPS
NAVSHIPS 250-660-35 - Manual for Design of Electrical Equipment with Small Stray Magnetic Field.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules and Regulations

(Application for copies should be addressed to Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

3. REQUIREMENTS

3.1 Diesel generator sets.-

3.1.1 The engine, generator, power take-off (when specified, see 6.1) and accessories shall be of lightweight construction. Nonmagnetic materials shall be employed as specified in 3.2.1 and 3.6. The complete unit shall be mounted on a common sub-base of aluminum construction. After installation in the ship, the unit may be mounted on sound isolation mounts; accordingly, the sub-base shall be suitable for support without further bracing or stiffening. Eyebolts or other means for lifting the complete unit shall be furnished. Suitable provision shall be made for temperature differentials between the engine and base.

3.1.1.1 The mechanically actuated, directly coupled power take-off (when specified) shall be mounted on the forward end of each engine. The power take-off shall have a short shaft and keyway in accordance with the latest Society of Automotive Engineers (SAE) standards and shall be capable of driving the connected hydraulic pumping units specified (see 6.1) at rated generator speed. A drive pulley or gear on the output shaft of the power take-off shall not be furnished. Except for bearings and clutch plates, all parts of the power take-off shall be nonmagnetic in accordance with 3.1.9.

3.1.2 Design.- The diesel generators shall be capable of satisfactory parallel operation in accordance with kilowatt load division and reactive load division requirements of MIL-G-3048 and MIL-R-2729, respectively.

3.1.3 The generator frame shall be rigidly connected to the flywheel housing by means of a bell housing.

3.1.4 The engine end of the generator shaft shall be piloted in the engine flywheel, or by other means, to maintain correct alignment of the armature and crankshaft.

3.1.5 Starting system.- A nonmagnetic hydraulic starting system shall be furnished complete in all respects and included as an integral part of the diesel engine.

3.1.6 Identification plate.- An identification plate of the type specified in MIL-G-3048 shall be furnished.

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3.1.7 Weight and dimensions.- The engine and generator unit shall not exceed the weight and space requirements, including space required for removal of generator and engine components, as specified (see 6.1).

3.1.8 The unit shall be so designed and constructed that under starting and all operating conditions, the unit will meet the stray field limit and requirements specified in confidential amendment-1.

3.1.8.1 All electrical equipment shall be designed to keep stray magnetic fields at a minimum by making the area of current loops as small as practicable and arranging adjacent loops so that their field will be in opposition. Current loops shall not enclose any magnetic material unless essential for proper performance of equipment.

3.1.8.2 Publication NAVSHIPS 250-660-35, should be used as a guide in designing the equipment to minimize the effect of stray fields:

(a) Two-pole machines should not be used.
(b) Solid magnetic frames should be used; that is, no split frames.
(c) The outside of the magnetic structure should be a smooth figure of revolution. Protuberances of magnetic material, such as feet or supports should be avoided. Pole bolts should be recessed, and, in general, every effort made to ensure that the outside of the magnetic structure is a smooth surface generated by revolving the arc of a curve about the axis of the machine.
(d) The number of interpoles should be the same as the number of main field poles.
(e) The number of turns in any field pole winding should be the same on each pole.
(f) All heavy current carrying conductors in the generator should be arranged with the greatest of care to avoid uncompensated current loops.
(g) The cross section of circular frame should be uniform in thickness all around to within plus 1.0 percent.

3.1.9 Nonmagnetic materials used in the engine, accessories, and the electrical equipment (see 3.2.1 and 3.6) shall be as approved by the bureau or agency concerned. The maximum relative permeability of nonmagnetic parts shall not exceed 2.0 after fabrication. If the supplier feels that a relative permeability requirement of 2.0 or less after fabrication cannot be met or is not feasible for any particular part or parts, he shall forward a detailed list of such parts (and reasons for noncompliance) at which time the bureau or agency concerned will decide whether a waiver can be granted.

3.1.9.1 Within 60 days from the date of the contract or order, a material list for the complete diesel generator set showing material used for each part shall be submitted to the bureau or agency concerned for approval.

3.1.10 Shock.- The complete generator set with all attached and detached accessories shall be designed to meet the requirements for grade A, type A, class I equipment of MIL-S-901 and as specified in 3.1.10.1 through 3.1.10.4.

3.1.10.1 Intended function.- The intended function of shockproof generator sets is to continuously deliver electrical power throughout and after application of shock in the magnitude experienced on shock test machines. Further, a shockproof generator set, when subjected to the above mentioned magnitude of shock shall not suffer damage to the extent that it creates a possible hazardous situation such as fire or injury to operating personnel or where shock damage would likely cause premature failure of major components (diesel generator and excitation system) requiring immediate shutdown of the generator set to prevent such failures.

3.1.10.2 Acceptance of equipment shall be based on ability to perform its intended function as defined in 3.1.10.1; the magnitude of damage revealed during the post shock test inspection at the supplier's plant and corrective action proposed by the supplier where inspection reveals damage.

3.1.10.3 Generator sets and detached accessories shall successfully complete the shock test specified in 4.2 herein.

3.1.10.4 In addition to the requirements in MIL-S-901, the following requirements apply in designing equipment for class HI (High Impact) shock:

(a) Equipment sub-base holding down bolt holes provided for securing equipment to the ship's structure shall be designed for use with nonfitted bolts conforming to grade 2, types I, II and

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III of MIL-B-857 and self locking nuts (MS17828, MS17829 and MS17830) conforming to MIL-N-25027. Maximum clearance between holding down bolt holes and bolts shall not exceed the following:

Nominal bolt diameter	Maximum diameter of hole
3/4 inch or smaller	Nominal bolt diameter plus 1/32 inch
Larger than 3/4 inch	Nominal bolt diameter plus 1/16 inch

(b) Pipe threaded connections.- Taper pipe threaded connections between components of the generator set (including detached items) shall not be used unless specifically approved by the Bureau of Ships. In general, pipe threaded connections shall not be used where failure of the connection could cause a major breakdown of the generator set or create a hazard to equipment located near the generator set (such as spraying the generator set or other equipment with oil or water) and where failure presents a hazard to operating personnel (such as a taper pipe threaded plug acting as a projectile upon failure under shock).

(c) The design may incorporate use of resilient mountings for instruments, gages, contactors and relays secured to the generator set package and for the diesel control panel. Resilient mounts shall conform to the requirements of MIL-M-17185.

3.1.11 In the construction of the engine-generator set and accessories, all screw threads tapped in aluminum to receive cap screws shall be protected by the installation of steel inserts.

3.1.12 The design of all electrical equipment shall be approved by the bureau or agency concerned prior to manufacture.

3.1.13 The complete diesel generator set shall be painted in accordance with MIL-E-917.

3.1.14 Vibration.- The combined diesel generator set shall meet the requirements of type I and III of MIL-STD-167.

3.2 Diesel engine.- The diesel engines shall be as specified herein.

3.2.1 The following major parts and accessories of the engine shall be made of nonmagnetic material:

(a) Cylinder block.
(b) Cylinder head.
(c) Flywheel housing.
(d) Oil cooler housing and cover.
(e) Jacket water cooler housing and cover.
(f) Water outlet manifold and piping.
(g) Exhaust manifold and flange.
(h) Oil pan.
(i) Air inlet silencer.
(j) Water pump inlet elbow.
(k) Thermostat housing.
(l) Heat exchanger tank.
(m) Vibration damper.
(n) Flywheel.
(o) Muffler.
(p) Cylinder liners.
(q) Connecting rods.
(r) All pumps and filters, except fuel injector pumps.
(s) Fuel and lube oil piping, access covers, miscellaneous hardware (not highly stressed), mounting bolts, brackets, miscellaneous covers, and gages.
(t) The camshaft and balance shaft front and rear balance weights.

3.2.2 Painting.- All external surfaces of ungalvanized ferrous metal nonworking parts, the operating temperature of which is 300° F. or less, shall be thoroughly cleaned and coated with one coat of pretreatment (formula No. 117 for metals) in accordance with MIL-P-15328, and two coats of equipment enamel, semigloss, class 2 of MIL-E-15090. Internal surfaces of the crankcase and oil pan shall not be painted or primed.

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(e) 10 minutes at minimum idling load

3.2.3 Rating.- The engine shall be capable of driving its generator to produce 100 kw. continuously and 110 kw. for 2 hours without injury to any part. The normal engine speed shall be 1800 revolutions per minute (r.p.m.).

3.2.3.1 Engines shall be of a type which have been in commercial or Navy service at the same or comparable rating.

3.2.4 Accessory equipment.- The engine shall be complete in all aspects and shall include power take-off (when specified), engine driven fresh water pump, positive displacement sea water pump, lubricating oil pressure pump, hydraulic starting equipment, overspeed trip, governor, strainers, filters, coolers, mufflers and all necessary instruments with gage board.

3.2.5 Smokeless operation.- The engine exhaust shall not be excessive when the engine is operating at 5 percent in excess of rated load and normal r.p.m. and operating with fuel oil in accordance with MIL-F-16884. Lubricating oil shall be in accordance with MIL-L-9000.

3.2.6 Torsional vibration.- Torsional vibration shall be in accordance with MIL-STD-167.

3.2.7 Inclination.- The engine shall be designed to operate in accordance with this specification and without loss of oil when inclined permanently from the normal horizontal position, up to 30 degrees to either side or 15 degrees on either end.

3.2.8 Lubrication system.- The engine sump shall be provided with an oil level indicator, filling opening, and an accessible drain connection or a hand pump to empty all accumulations of oil. The lubrication of the power take-off (when specified) shall be independent of the engine lubrication system. Parts of the power take-off requiring lubrication may be lubricated by grease or lubricating oil.

3.2.8.1 Lubricating oil filters shall be provided. Full flow filter elements shall conform to MIL-F-20707. Cases for full flow filters shall be specifically approved by the bureau or agency concerned.

3.2.8.2 An oil cooler, using fresh water as cooling medium, shall be supplied for each engine. The cooler shall be in accordance with either MIL-C-15730 or MIL-C-21121 except that hose connections are acceptable. Plate or plug-in core type coolers are acceptable.

3.2.9 Cooling systems.- The engine shall be provided with a heat exchanger for cooling the circulating water, using sea water as cooling medium. The cooler shall be in accordance with MIL-C-15730 or MIL-C-21121, except that hose connections are acceptable. Plate or plug-in core type coolers are acceptable.

3.2.9.1 Temperature control.- The temperature of the engine circulating water shall be controlled automatically by means of a thermostatic valve.

3.2.9.2 Operating temperature.- The fresh and sea water pumps, coolers and thermostatic valve shall be of ample capacity to maintain the lubricating oil and cooling water temperatures within the following range:

(a) Sea water inlet temperature: Maximum 85°F., minimum 35°F.
(b) Lubricating oil operating temperature: Maximum 230°F., minimum 140°F.
(c) Cooling water from engines: Maximum 180°F., minimum 150°F.

3.2.9.3 Water pumps.- Fresh water pumps shall be of the centrifugal type. Sea water pumps shall be of the positive displacement type.

3.2.9.4 Drainage.- Means for complete drainage of the engine cooling water and sea water systems shall be provided.

3.2.10 Fuel system.- The engine fuel system shall include a positive displacement rotary type transfer pump, by-pass relief valve, and filter. The filter shall be in accordance with MIL-F-20627 and when practical shall be installed between the transfer pump discharge and injectors. The fuel strainer shall be the single unit metal edged type, with 0.003 inch spacing incorporating enclosed manual cleaning knives and a sediment drain valve. The strainer case shall be of brass or bronze. Inlet and outlet connections shall be 3/8-inch iron pipe size (i.p.s.).

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3.2.11 Regulating governor.- The engines shall be provided with a governor incorporating a speed-drop mechanism, adjustable from 0.5 to 5 percent of rated speed. The governor shall be capable of holding the engine speed constant to within plus or minus 0.5 percent of operating speed for any constant value of load between 10 percent load and full load. The response of the governor shall be such that upon sudden change from no load to full load and full load to no load, the maximum speed deviation from normal shall not exceed 7 percent. In addition, the speed shall return to and remain within 1 percent of the final steady-state speed in not more than 5 seconds following the change in load. The governor shall incorporate a 115 volt a.c., speed changing motor to provide for remote speed control of the engines.

3.2.11.1 Overspeed governor.- The engine shall be provided with a manual resetting overspeed governor, of the trip type, which will shut off fuel or air to the engine when the speed exceeds 115 percent of normal operating speed.

3.2.12 Starting system.- A hydraulic starting system shall consist of a hydraulic starter motor and all accumulators, reservoirs, control valves, filters, pressure regulators, pressure gages and pumps necessary to make the system self-contained and non-dependent on any other ship system. This starter system shall be properly adapted to the engine and of adequate capacity to provide 5 consecutive starts from one full pressure accumulation of the starting system oil (pump inoperative). An electric motor driven pump, automatically controlled by an accumulator pressure activated switch shall be provided to maintain the accumulator pressure at some usable pre-set level. In addition, a hand operated pump shall be provided for use during cold ship start-up or emergencies. It shall be practical for one operator to pump up full pressure (from zero pressure) within 5 minutes. A push button activated solenoid start-up control valve shall be furnished. Provision shall also be made for manually activating the control valve in event of loss of power or solenoid failure. The electric pump motor and solenoid valves shall be suitable for use from a 115 volt, single phase, 60 cycle a.c. power source.

3.2.13 Air silencer.- The air intake to the engine shall be equipped with an effective air intake silencer and screen.

3.2.14 Exhaust system.- The exhaust manifold shall be water cooled and provided with flange connection and a straight threaded companion flange. An air cooler is required for turbocharged engines.

3.2.14.1 Exhaust muffler.- A muffler in accordance with type A of MIL-M-15337 shall be furnished.

3.2.15 Alarm connections.- The supplier shall provide a 3/8-inch i.p.s. thread connection at the remote end of the lubricating oil distribution system, for later installation of a low lubrication oil pressure alarm contact maker. Space for installing a high temperature contact maker actuating bulb 4-1/2 inches long with 1/2 inch i.p.s. thread, in the cooling water outlet header from the engine shall be provided. The holes shall be closed with pipe plugs. The holes shall be satisfactory for use with alarm switches in accordance with MIL-S-16032.

3.2.16 Instruments and gages.- Engines shall be equipped with instruments specified in 3.2.16.1 and 3.2.16.3. Gages or thermometers which are mounted on the gage board shall be of the flush mounted type. No stem mounted gages are permitted.

3.2.16.1 The following instruments and gages shall be furnished:

(a) Oil pressure gage.
(b) Water temperature gage.

3.2.16.2 An electric type tachometer in accordance with type IC/EFB of MIL-T-16049 shall be furnished. The scale range of all tachometers shall be at least 25 percent greater than engine rated speed.

3.2.16.3 Gage board.- All instruments shall be installed on a gage board mounted on the engine near the controls. The board shall be secured by means of vibration and shock isolating fittings to the engine.

3.2.17 Remote shutdown.- Provision for attachment of a remote mechanical shutdown device by the shipbuilder shall be made on the engine. The means provided shall shut off fuel or air to the engine, and shall be suitable for connection to flexible shafting.

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3.3 Generators.- Generators shall conform to MIL-G-3124 and MIL-E-917 except as otherwise specified herein:

3.3.1 The generator shall have the following classification:

Type	II alternating current
Primer mover	Diesel engine
Rating:	
Voltage	450 volts a.c.
Kilowatt capacity	100 kw. continuous
Power factor	0.8 lagging
Phases	3
Speed	1800 r.p.m.
Enclosure	Dripproof, protected
Type of bearing	Sleeve or ball (see 3.3.2)
Ambient temperature	50° Centigrade (C.)
Insulation	Class B, F, H or N
Parallel operation	Required
Starting	Hydraulic
Overload	150 percent current for 5 minutes

3.3.2 If a ball bearing is employed, it should be a double shielding type in accordance with FF-B-171 with grease lubrication. Adequate grease seals shall be provided and the lubrication system so designed that grease will not leak into the generator under any operating conditions.

3.3.3 Electrical insulation.- Insulation systems shall be class B, F, H or N, and shall conform to MIL-E-917. The number of impregnations of each type of generator shall conform to the minimum requirements of the manufacturer's detail insulation drawings as approved by the Bureau or agency concerned for that specific type generator. In any case, there shall be not less than three complete and independent impregnations of any electrical winding. One impregnation shall be defined as one complete cycle of treatment and curing (such as one immersion and one baking) by a recognized method. However, the brushing or spraying method shall not be considered as one of the required impregnations unless it is applied as the final coat of varnish. All insulated windings shall receive at least two of the required impregnations after the windings have been installed on the corresponding machine subassemblies. Except as otherwise specified herein, all insulated electrical windings shall receive at least one of the required impregnations before assembly on machine parts. Impregnation before assembly on machine parts will not be required for windings which are to be inserted in semi-enclosed slots or other windings where the method of installation or the insulation system used is such as to make this preliminary impregnation impractical. In such cases, all required impregnations shall be made after the windings are installed on the machine parts. The use of solventless impregnating varnish may preclude the use of a subsequent varnish dip and bake and shall be subject to Bureau or agency approval.

3.3.4 Painting.-

3.3.4.1 Exterior.- Except for shafts and identification plates, exterior surfaces of the generator shall be painted in accordance with MIL-E-917.

3.3.4.2 Interior.- Electrical insulation, surfaces in contact with lubricating oil or grease, commutators, collector rings, brushes, bearings, and bearing surfaces shall not be painted. Also, peripheries of rotors and any other rotating part of the machine from which centrifugal force may cause the paint to be thrown on the winding when the machine is operating at rated load and speed and at normal temperature, shall not be painted; insulation varnish conforming to MIL-V-1137 may be applied to such parts. All other parts that are not corrosion-resisting (see MIL-E-917 for listing of corrosion-resisting materials and treatment) shall be painted in accordance with MIL-E-917 except that one coat of enamel of light color will be sufficient.

3.3.5 Generator capacity rating.- The kilowatt rating shall be the net power available for continuous use. If the generator has any auxiliary equipment, such as a static excitation system that receives power from the generator and is considered to function as an essential part of the basic generator set, the generator shall have capacity in excess of the new power rating to furnish this power. Such capacity is considered to be the gross kilowatt rating.

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3.3.6 Efficiency.- The generator efficiency shall be not less than 89.5 percent at rated load.

3.3.7 Wave shape.-

3.3.7.1 Deviation factor.- The deviation factor of the open circuit terminal voltage shall not exceed 6 percent on any or all phases.

3.3.7.2 Harmonic content.- From no load to full load, the root mean square (r.m.s.) value of the sum of all of the harmonic components in the output voltage wave shall not exceed 5 percent of the r.m.s. value of the rated output voltage. No single harmonic shall exceed 3 percent. Harmonics of 0.2 percent or less of the r.m.s. value of the rated output voltage may be disregarded in the determination of the harmonic content of the output voltage wave.

3.4 Excitation and voltage regulation system.- The excitation and voltage regulation system shall be of the completely static type or rotary amplifier type in accordance with type III or type II of MIL-R-2729.

3.4.1 Static exciter.- The components (except for the power rectifiers) of the static exciter shall be enclosed in a dripproof, convection cooled enclosure suitable for deck and bulkhead mounting. The power rectifiers shall be enclosed in a dripproof enclosure designed for mounting in and cooling by the generator air-intake stream. Consideration will be given to the use of a permanent-magnet pilot exciter for flashing the field. The exciter and field flashing device shall require no source of current external to the set being excited.

3.4.1.1 The manufacturer's proposed method of excitation shall be submitted to the bureau or agency concerned for approval prior to manufacture.

3.4.2 Voltage regulator.- The voltage regulator shall be in accordance with type II or type III of MIL-R-2729 and shall have the following classification:

Type of control	Static
Enclosure	Switchboard
Ambient temperature	50°C.
Shock classification	Class HI (high impact)
Operation	Parallel
Regulators per switchboard	One normal

3.4.2.1 All voltage regulator components and parts except the manual control unit, the voltage adjusting unit, the cutout switch, and the current droop compensation transformers, shall be mounted in one enclosure (box) suitable for mounting in the switchboard framework. The dimensions of the enclosure shall not exceed 22-1/2 inches in length, 15 inches in width, and 14 inches in depth. The components mounted in the enclosure shall be readily serviceable from the front of the switchboard by employing a hinged or screw type cover removable from the front. The manual control unit, the voltage adjusting unit, and the cutout switch shall be suitable for back of board mounting (operable from front of panel).

3.4.3 Semiconductor devices shall be selected from MIL-STD-701.

3.5 Shock.- The equipment shall be designed to meet the HI shock requirements of MIL-S-901.

3.6 Nonmagnetic material.- In order to keep the magnetic signature of the equipment at a minimum, nonmagnetic material shall be employed in the construction of the following parts:

Generator

Shaft (where not used in the magnetic circuit)
End frames
Ventilating covers
Fan and driving disk assembly
Collector ring shell
Brushholder studs
Mounting feet
Frame (where not used in the magnetic circuit)
All nuts and bolts, lifting eyes and miscellaneous hardware

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Static exciter regulator and rheostats

All enclosures and covers
All mounting hardware, actuating shafts, handles, identification plates and nuts and bolts.

Starting motor

End bells
Bendix drive housing
Shaft
Nuts and bolts and miscellaneous hardware

Note.- In the event that use of nonmagnetic material for any component item would preclude compliance with any of the specification requirements, the matter shall be referred to the bureau or agency concerned for approval.

3.7 Electromagnetic interference.- The design of the generator shall be such as to minimize the generation of electromagnetic interference voltage over the frequency spectrum specified in MIL-I-16910 without addition of capacitors or filters. The limits specified in MIL-I-16910 shall be used as a design target. Radiated interference tests and conducted interference tests in accordance with MIL-I-16910 shall be conducted when required (see 6.1).

3.8 Electrical drawings.- Submission and approval of generator and regulator drawings shall be in accordance with the procedures outlined in MIL-D-963. The generator and excitation voltage regulation system drawings shall contain the information as required by MIL-G-3124 and MIL-R-2729, respectively. An installation drawing for the complete set shall be furnished and shall conform to the requirements of MIL-G-3048. All installation data required by MIL-G-3048 shall be on one drawing. Pertinent contractual information shall appear on the installation drawing. All drawings shall be submitted for approval within 60 days after date of contract or order.

3.9 Engine drawings.- Complete working drawings of the engines in the form of microfilm or blackline prints suitable for microfilming shall be furnished. One set of detail drawings shall be furnished to Ships Parts Control Center, Mechanicsburg, Pennsylvania for assignment of Federal stock numbers in accordance with MIL-P-15137, if Federal stock numbers have not been assigned on prior contracts.

3.10 Drawings for shipbuilding activities.-

3.10.1 Two blueprint copies of all outline and assembly drawings of the complete unit and major components (such as generator, exciter, and regulator) and unattached accessories shall be forwarded to shipbuilding activities as designated by the bureau or agency concerned at the same time (and each time) they are submitted to the bureau or agency for approval. These drawings shall be stamped preliminary until final approval action has been taken.

3.10.2 One complete set of reverse reading vandykes of all applicable mechanical and electrical detail and component drawings shall be furnished to a shipbuilding activity designated by the bureau or agency concerned.

3.11 Manuals.- Manuals, including parts lists, shall be comparable to type III of MIL-M-15071. The manual shall be submitted to the bureau or agency concerned for approval prior to shipment. A manual consisting of the following material shall be considered as complying with type III:

(a) The manufacturer's standard engine manual, modified to indicate applicable part numbers for any modified engine parts, and including under the "maintenance" section a table showing normal and permissible limits of clearances, diameters and thicknesses for new wearing parts. Under the section "operation," or as part of the page "engine specifications," a table giving normal full load temperatures and pressures, and permissible limits shall be included. Torque settings of important nuts and bolts shall also be specified.
(b) A reduced size copy of the combined unit installation drawing.
(c) Photographs of the unit fore and aft.
(d) A comprehensive electrical insert, including reduced size copies of all finally approved electrical drawings.

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3.11.1 If offset printing is used in the publication of the manual, a complete set of negatives shall, after completion of the manual, be furnished to the bureau or agency concerned and shall remain the property of the Government. Regardless of the method of printing used, one glossy print negative of each halftone illustration included in the manual shall be furnished the bureau or agency concerned.

3.12 Repair parts.-

3.12.1 Electrical repair parts and tools shall be furnished in accordance with the individual equipment specification.

3.12.2 Mechanical repair parts and special tools, where applicable, shall be furnished with each engine. Special tools are defined as those tools not listed in the Federal Supply Catalog. (Copies of this catalog may be consulted in the office of the Government inspector.)

Connecting rod	1
Piston complete with pin, bushings and locks	1
Cylinder liner	1
Connecting rod bearing shells (upper and lower)	2
Main bearing shells (each type and size, upper and lower)	1
Bushings (each type and size)	1
Ball and roller bearings	1
Push rod assembly and tappets	2
Exhaust valves	4
Intake valves	2
Valve guides	4
Valve springs, keys and locks	20 percent
Piston rings	100 percent
Fuel injection nozzle assembly or combined unit fuel injection pump and nozzle assembly	50 percent
Unit fuel injection pump assembly for type where injector is not integral with pump	15 percent
Spray tips	100 percent
Nozzle holders	2
Thermostatic valve	1
Electrolysis protectors (cooler zincs)	100 percent
Lube oil filter elements	300 percent
Fuel oil filter elements	100 percent
Seawater pump impeller	1
Fresh water pump impeller	1
Hose connections	100 percent
V-belts	100 percent
Gaskets, packing O-rings and seals	100 percent
Springs (each type and size)	1
Coupling spring packs	50 percent
Roller or silent chain	100 percent
Special tools	1 set

Percentages expressed are that portion of the sum total of that item used in the construction of one engine. In addition to the parts outlined, the supplier shall include those repair parts for other items requiring replacement during engine maintenance within the first year of operation. Items may be deleted or substituted as applicable to the particular engine furnished. Repair parts lists based on the above quantities shall be prepared and submitted in accordance with MIL-P-15137. The list of parts shall be made up in specific groups of related or component parts. Each assembly or group shall list all parts associated with the "set," the supplier shall list the various tools which comprise a "set."

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable

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sea water pump shall not be 4.4.2 During tests, the temperature of the cooling water supplied by the

to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Shock tests.- Shock tests shall be conducted in accordance with the following (see 6.2):

4.2.1 Generator sets or detached accessories which are within the weight and space capacity of shock testing machines shall be subjected to and successfully pass the shock tests. Where a manufacturer has conducted shock tests in accordance with this specification and the equipment has been accepted by the Bureau of Ships, a test report of the previous test should be forwarded to the Bureau of Ships for extension approval.

4.2.2 Shock tests on the generator set package and detached accessories shall be conducted in accordance with the grade A machinery requirements and type A tests of MIL-S-901 and the following:

(a) The generator set package (complete with diesel generator, exciter if attached, sub-base and all attached components and accessories) and any detached components needed to control operation of the diesel shall be shock tested as a principal unit. Other detached components may be tested along with the generator set package if the weight and size limitations of the shock testing machine are not exceeded. Where these other detached components are not tested along with the generator set package, they shall be tested as separate items using the applicable shock testing machine referenced in MIL-S-901 and in accordance with the requirements in the individual equipment specification covering each detached component.

4.2.3 Operation of equipment and instrumentation during shock tests shall be as follows:

(a) Tests on the generator set package and any detached component needed to control operation of the diesel and where other detached components are tested along with the generator set package. Equipment shall not be operated or energized during each blow or shot except for the last blow or shot required by MIL-S-901. After each blow or shot, the equipment shall be inspected for damage. Correction of minor damage during the progress of tests will be permitted. Where no damage has been experienced, or after minor damage has been corrected, the equipment shall be operated before proceeding to the next blow or shot. Major damage shall be cause for rejection and will require retest at the test level which caused the damage before proceeding to the next blow or shot. During the last blow or shot required by MIL-S-901, equipment shall be energized and operated at no load, rated voltage and rated speed. During this blow or shot an oscillograph recording shall be made showing generator output voltage and frequency characteristics from a time beginning with 5 seconds before the blow or shot until 30 seconds after the blow or shot.
(b) Where detached components, other than those required for control or diesel operation, are not tested along with the generator set package, equipment shall be operated or energized and instrumentation shall be as required by the individual equipment specification.

4.2.4 Mounting of equipment for shock tests shall be as follows:

(a) Light weight equipment.- The generator set package and, unless otherwise specified in the individual equipment specifications, detached accessories or components shall pass shock tests when mounted on fixtures 4A or 4C of MIL-S-901, as appropriate.
(b) Medium weight equipment.- The generator set package and, unless otherwise specified in the individual equipment specifications, detached accessories or components shall pass shock tests when mounted on fixtures shown on figures 10-1 and 10-2 of MIL-S-901, as appropriate.

4.2.5 Post shock test inspection and reports.- After completion of shock tests, equipment shall be returned to the supplier's plant for complete teardown and inspection for damage. In addition to checking for damage to individual components, the generator set sub-base shall be checked for damage (including distortion) and a check shall be made to insure that shock tests did not result in misalignment of components on the sub-base. Where damage is minor and is not of a nature which could result in major failure of components during continuous operation, the equipment is considered acceptable from a shock standpoint. The supplier shall forward a list showing identification of components and parts inspected and details of damage discovered during the post shock test inspection to the Bureau of Ships for review and comment. Where damage is of a nature which would result in major damage to components under continuous operation, the supplier shall also forward information on proposed modifications to eliminate the deficiency. Only the Bureau of Ships will determine the acceptability of a unit and all reports should be forwarded for approval.

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4.2.6 Disposition of shock tested equipment.- Unless otherwise specified in individual equipment specifications, shock tested equipment will be acceptable for delivery under a contract or order provided the following requirements are met:

(a) The post shock test inspection reveals no damage or where damage is discovered it can be corrected.
(b) All damage resulting from shock tests is corrected. In this regard, all bearings shall be replaced where inspection reveals physical signs of minor damage (such as slightly marked surfaces) due to shock tests. Where major damage to bearings is discovered, appropriate redesign corrective action shall be taken.
(c) Any redesign indicated as the result of shock tests is incorporated in the shock tested unit.
(d) After all damage has been corrected and all new design features have been incorporated in the shock tested equipment, the assembled equipment shall be subjected to the routine tests required by MIL-G-3048 and a four hour continuous load test. In addition, electrical equipment shall be subjected to and successfully pass the insulation resistance and dielectric tests required by individual equipment specifications after completion of shock tests. Results of these tests shall show conformance with the performance requirements of MIL-G-3048 and individual equipment specifications. Further, these results shall be compatible with the results of tests taken on other equipment of the same design and type and results shall show that operation is normal in all respects.

4.3 The requirements of MIL-G-3124 for obtaining the generator shut down temperature by extrapolating the temperature-time curve back to zero time shall not apply. In lieu thereof, the maximum observable temperature obtained shall be considered as the shut down temperature. All other requirements for method of measuring temperature shall apply.

4.4 First production engine performance tests.- Test shall be made only on the first application of a new engine by the bureau or agency concerned. Where a manufacturer has conducted first production engine performance test in accordance with the requirements of this specification and the equipment has been accepted by the Bureau of Ships, a test report of the previous test will be acceptable in lieu of the actual test.

4.4.1 Endurance test.- The first engine shall be run for 125 eight hour cycles as follows:

Two hours at rated load and speed.
One hour at 85 percent load and rated speed.
Ten minutes at minimum idling load and speed.
One hour and 50 minutes at rated load and speed.
Ten minutes at minimum idling load and speed.
Thirty minutes at 85 percent load and rated speed.
Ten minutes at minimum idling load and speed.
Ten minutes at 85 percent load and rated speed.
One hour and 50 minutes at 110 percent rated load and rated speed.
Ten minute shut down.

4.4.2 During tests, the temperature of the cooling water supplied by the sea water pump shall not be less than 85°F. The engine shall be tested with all attached accessories furnished with the engine, including coolers and temperature regulating valve.

4.4.3 The exhaust back pressure at the engine manifold outlet shall be set at 3 inches of Hg at rated load and speed by placing a restriction in the exhaust line. The restriction set for full power and speed shall remain fixed during all runs of the first unit engine tests.

4.4.4 Fuel used during the tests shall conform to MIL-F-16884.

4.4.5 Lubricating oil used during the tests shall conform to MIL-L-9000.

4.4.6 The following data shall be recorded at regular intervals. When a phase of a test is of such short duration that operating temperatures cannot be stabilized, no reading is required.

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conformance with level A of MIL-E-17555 and MIL-P-16232, as applicable.

(a) Time.
(b) Ambient temperature.
(c) Fuel consumption.
(d) Lubricating oil pressure.
(e) Lubricating oil temperature (sump).
(f) Sea water temperature in.
(g) Sea water temperature out.
(h) Fresh water temperature out of engine.
(i) Fresh water temperature in engine (at pump inlet).
(j) Exhaust temperature (at manifold outlet).
(k) Exhaust back pressure.
(l) Engine speed.
(m) Generator output (kw., volts, amperes, power factor).

4.4.7 Post trial examination.- After completion of the endurance test, the engine shall be disassembled and inspected for the condition of all parts. All parts shall be in satisfactory operating condition without excessive wear or distress on any parts of the engine. A description of the condition of the parts shall be included in the report of tests.

4.5 Combined unit tests.- Only one unit shall be subjected to the periodic test as specified in 4.5.1 herein.

4.5.1 First unit test.- The diesel generator set shall be subjected to the periodic tests of MIL-G-3048 except as specified in 4.5.1.1 and 4.5.1.2.

4.5.1.1 The governor tests shall not be conducted. In lieu thereof, sufficient tests shall be conducted to insure conformance with 3.2.11 as determined by standard strip charts. The setting of the overspeed governor shall be determined by overspeeding the engine and recording maximum speed, at which the trip functions, or the strip charts.

4.5.1.2 Starting tests shall be conducted under temperature conditions specified in MIL-G-3048 to show conformance with 3.2.12 herein.

4.5.2 Sampling for selective tests.- Engine generator sets shall be selected in accordance with table I. If the routine tests (sec 4.5.3) show large variations from the accepted design and performance, tests on additional sets may be required.

Table I - Sampling for selective tests.

Number of engine generator sets on contract or order	Minimum number of sets to be tested
2 to 4	2
5 to 8	3
9 to 15	4
16 to 25	5
26 to 40	6
41 to 65	8
66 to 110	10
111 to 180	13

4.5.2.1 Selective tests.- Each engine generator set selected in accordance with 4.5.2 shall be subjected to the tests specified in MIL-G-3048 as selective tests. If any set fails in one or more tests no deliveries shall be permitted until the manufacturer has accomplished the necessary correction and has demonstrated it in two sets. The same change shall be made in all sets which have not been delivered.

4.5.3 Duplicate unit tests.- All duplicate units shall be subjected to the routine tests specified in MIL-G-3048, except that the continuous test at full load shall be run for 4 hours in lieu of 1 hour. Also, two starts shall be made with the hydraulic starting system.

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5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.-

5.1.1 Diesel generator set.-

5.1.1.1 Preservation and packaging.- Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.2 Packing.- Packing shall be accomplished in a manner which will insure acceptance by common carrier at lowest rate, and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.3 Marking.- Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the supplier's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, manufacturer's name and destination.

5.1.2 Onboard repair parts.- Onboard repair parts shall be processed as follows:

5.1.2.1 Mechanical repair parts.- Mechanical repair parts shall be preserved and packaged level A, packed level C and marked levels A and C, respectively in accordance with MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment.- The requirements, and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Preservation and packaging.-

5.2.1.1.1 Level A.-

5.2.1.1.1.1 Diesel generator sets.- Diesel generator sets shall be preserved and packaged in accordance with type 4 or type 5, method 1 or 2A as specified, of MIL-P-16298 or MIL-E-17555, as applicable. When the size and mounting arrangement of the generator set does not permit enclosure of the generator within a water vaporproof barrier as required in a method II package, the alternate method of packaging specified in MIL-P-16298 may be applied.

5.2.1.1.1.2 Repair parts.- Repair parts shall be processed as follows:

5.2.1.1.1.2.1 Mechanical repair parts.- Mechanical repair parts shall be preserved and packaged in accordance with MIL-R-196.

5.2.1.1.1.2.2 Electrical repair parts.- Electrical repair parts shall be preserved and packaged in accordance with level A of MIL-E-17555 and MIL-P-16298, as applicable.

5.2.1.1.2 Level C.- Preservation and packaging of turbine generator sets shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

5.2.1.2 Packing.-

5.2.1.2.1 Diesel generator sets.-

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5.2.1.2.1.1 Assembled sets.-

5.2.1.2.1.1.1 Levels A and B.- When the package set is shipped as a complete package, each set shall be anchored, blocked, braced and cushioned and enclosed in a sheathed crate conforming to MIL-C-104 and appendix thereto.

5.2.1.3 Marking.- In addition to any special marking requirements, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.2.1.3.1 Special marking.- Shipping containers shall be stencilled as follows:

CAUTION - THIS EQUIPMENT MAY BE DAMAGED
UNLESS UNPACKING INSTRUCTIONS ARE CAREFULLY FOLLOWED; UNPACKING INSTRUCTIONS
ARE LOCATED (state where located).

When practical, this marking shall be applied adjacent to the identification marking on the side of the container.

5.2.1.3.2 Unpacking instructions.- In addition to any special marking required by the contract or order, unpacking instructions of equipment shall be provided to prevent possible damage during removal. When practical, one set of these instructions shall be placed in a sealed waterproof envelope prominently marked "Unpacking Instructions" and securely affixed to the outside of the shipping container in a protected location. If the instructions cover a set of equipment packed in multiple containers, the instructions shall be affixed to the number one container of the set.

5.2.2 Repair parts.- Repair parts shall be packed level A or B in accordance with the applicable packaging specification (see 5.2.1.1.1.2). Unless otherwise specified in the contract or order, repair parts shall be packed separately and shipped concurrently with the basic equipment.

5.2.3 Manuals.- Manuals shall be packaged and packed in accordance with MIL-M-15071.

5.2.4 Drawings.- Drawings shall be packaged and packed for the level specified in accordance with the drawing specifications specified in 3.8.)

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Power take-off, if required (see 3.1.1).
(c) Horsepower required for driving hydraulic pumps (see 3.1.1.1).
(d) Weight, length, width, and height (see 3.1.7).
(e) Stray field limit requirements (see 3.1.8).
(f) Whether radiated or conducted interference tests are required (see 3.7).
(g) Preservation, packaging, packing or marking requirements other than those required by 5.1 (see 5.2).

6.2 The following should be considered in connection with the shock tests:

(a) Preservation, packaging, packing and shipment of equipment to the shock testing activity.
(b) Any repair or redesign resulting from shock tests.
(c) Post shock test tear down, inspection, repair, modification and factory testing of the shock tested equipment at the supplier's plant.
(d) Furnishing of representatives to supervise installation and operation of equipment at the shock testing activity.
(e) Furnishing to the shock testing activity, an outline drawing or drawings of equipment showing overall dimensions; location and size of service connections, and arrangement and size of hold-down bolts, the weight and approximate location of the center of gravity.

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(f) Furnishing to the test activity all pertinent information regarding the method of operating the equipment during the test and what services (electric power, air, cooling water, and so forth) will be required for such operation.

6.3 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - SH
(Project 6115-N136SH)

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RETAINER SHEET

MIL-G-22997A(SHIPS)

MILITARY SPECIFICATION

GENERATOR SET, DIESEL ENGINE

100 KW, 450-VOLT A.C., 60-CYCLE, 1800 R.P.M.

Amendment - 1 to this specification contains information of a CONFIDENTIAL nature and is issued only upon specific request to the Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. When requesting copies, state the purpose for which required.

Preparing activity:
Navy - SH

FSC 6115

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 68 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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BUREAU OF SHIPS
WASHINGTON, D. C. 20360

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